Thermodynamic State Vector Discrepancy Mapping Iterator: Enforcing Conservation Laws in Biogeochemical Stock-Flow Simulations

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Sprint 076 – Module: `src/thermodynamics/state_validator.ts`

Abstract

Simulating complex planetary ecosystems and biogeochemical cycles requires rigorous adherence to foundational thermodynamic laws. In this report, we detail the architectural design, mathematical formalization, and TypeScript implementation of Sprint 076: the *Thermodynamic State Vector Discrepancy Mapping Iterator* (`src/thermodynamics/state_validator.ts`). By defining immutable, pure-functional mapping operators over elemental stock collections (Carbon, Nitrogen, Phosphorus, and Water), our system evaluates First Law conservation boundaries and Second Law entropy generation constraints in real time. We present the mathematical foundations of stock transfer discrepancies, the functional mapping contract, and empirical verification strategies demonstrating robust state validation across closed and open ecological systems.

1 Introduction & Systems Ecology Context

The *Web of Life* simulation engine models global nutrient cycles and energy flows across diverse trophic networks and geochemical compartments. As ecological simulations scale in temporal depth and structural complexity, numerical drift and unmonitored mass-energy imbalances can destabilize planetary models.

To prevent thermodynamic violations, Sprint 076 introduces `StateValidator`, a dedicated state validation subsystem. Operating within the monadic execution pipeline (`src/thermodynamic_monad_proc` this module maps current stock collections against strict conservation baselines, immediately isolating elemental discrepancies and ensuring that energy-mass conservation remains absolute.

2 Thermodynamic Foundations

2.1 First Law Conservation of Mass-Energy

The global state vector V tracks elemental species $e \in \{C, N, P, H_2O\}$. For any discrete simulation step, the conservation requirement dictates that net inflows minus outflows equal internal accumulation plus dissipated energy/matter:

$$\sum \Delta M_{\text{in}} - \sum \Delta M_{\text{out}} = \Delta M_{\text{internal}} + \Delta M_{\text{dissipated}} \tag{1}$$

The absolute discrepancy δ_e for any elemental stock relative to its expected baseline $M_{\text{expected},e}$ is quantified as:

$$\delta_e = |M_{\text{actual},e} - M_{\text{expected},e}| \tag{2}$$

2.2 Second Law & Irreversible Dissipation

In alignment with non-equilibrium thermodynamics, entropy generation $\Delta S_{\text{gen}} \geq 0$ must be maintained across state transitions. The discrepancy iterator flags anomalous states where conservation boundaries are breached beyond a predefined tolerance threshold ϵ.

3 Architectural Design & Implementation

The module is implemented as an immutable, pure-functional utility class `StateValidator` adhering to the `IStateValidator` interface.

```
export interface DiscrepancyRecord {
  element: 'C' | 'N' | 'P' | 'H2O';
  expected: number;
  actual: number;
  discrepancy: number;
  timestamp: number;
  isWithinTolerance: boolean;
}

export interface DiscrepancySummary {
  totalRecords: number;
  maxDiscrepancy: number;
  conserved: boolean;
  records: DiscrepancyRecord[];
}
```

3.1 Core Mapping Algorithm

The mapping function $\mathcal{M}(S, B, \epsilon)$ iterates over stock collections S and baseline vectors B to compute deviation metrics and generate immutable audit records.

4 Verification and Testing

Unit testing (`tests/sprint_076.test.ts`) verifies three primary operational regimes:

1. **Zero-Discrepancy Baseline:** Identical stock and baseline maps yield `conserved: true` with zero maximum deviation.
2. **Threshold Violation Injection:** Introducing an artificial perturbation ($\Delta C = 0.5 > \epsilon$) correctly triggers `isWithinTolerance: false`.
3. **Multi-Element Synchronous Aggregation:** Simultaneous validation across carbon, nitrogen, phosphorus, and water pools without performance degradation.

5 Conclusion

Sprint 076 establishes a rigorous mathematical and programmatic foundation for thermodynamic tracking within the *Web of Life* engine. By enforcing continuous state validation via functional iterators, the simulation architecture guarantees fidelity to physical conservation laws across all scale levels.

Repository Access: `https://github.com/pascalranoroarijaona/WebOfLife`